

Assignment 4
ISAN 5355 with Dr. Zhao
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1) 'book', 'branch', and 'branch_has_book' tables (in order)

The tables use the utf8mb4 character set by default and utf8mb4_0900_ai_ci for collation.

Column	Type	Default Value	Nullable	Character Set	Collation	Privileges	Extra
bookCode	char(4)		NO	utf8mb4	utf8mb4_0900_ai_ci	select,insert,update,references	
title	varchar(40)		YES	utf8mb4	utf8mb4_0900_ai_ci	select,insert,update,references	
publisherCode	char(3)		YES	utf8mb4	utf8mb4_0900_ai_ci	select,insert,update,references	
bookType	char(3)		YES	utf8mb4	utf8mb4_0900_ai_ci	select,insert,update,references	
paperback	char(1)		YES	utf8mb4	utf8mb4_0900_ai_ci	select,insert,update,references	

Column	Type	Default Value	Nullable	Character Set	Collation	Privileges	Extra
branchNum	decimal(2,0)		NO	utf8mb4	utf8mb4_0900_ai_ci	select,insert,update,references	
branchName	varchar(50)		YES	utf8mb4	utf8mb4_0900_ai_ci	select,insert,update,references	
branchLocation	varchar(50)		YES	utf8mb4	utf8mb4_0900_ai_ci	select,insert,update,references	

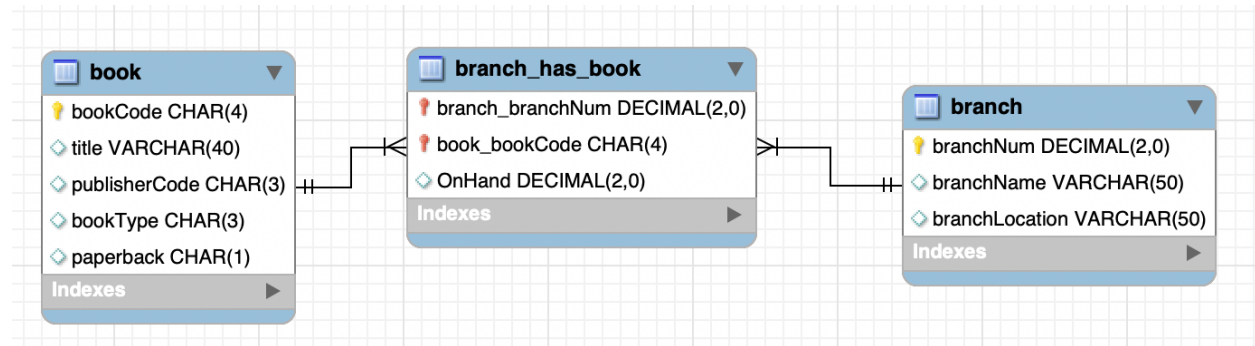
Column	Type	Default Value	Nullable	Character Set	Collation	Privileges	Extra
branch_branchNum	decimal(2,0)		NO	utf8mb4	utf8mb4_0900_ai_ci	select,insert,update,references	
book_bookCode	char(4)		NO	utf8mb4	utf8mb4_0900_ai_ci	select,insert,update,references	
OnHand	decimal(2,0)		YES			select,insert,update,references	

The tables use the InnoDB engine, done by setting the session default engine:
SET SESSION default_storage_engine = InnoDB;

```
25 -- Source - https://stackoverflow.com/a
26 -- Posted by Jocker, modified by community. See post 'Timeline' for change history
27 -- Retrieved 2025-11-11, License - CC BY-SA 3.0
28
29 • SELECT TABLE_NAME,
30       ENGINE
31 FROM   information_schema.TABLES
32 WHERE  TABLE_SCHEMA = 'assign4';
33
```

TABLE_NAME	ENGINE
book	InnoDB
branch	InnoDB
branch_has_book	InnoDB

2) ERR Diagram of new 'assign4' database



bookCode is the primary key for the 'book' table. *branchNum* is the primary key for the 'branch' table. *branch_branchNum* and *book_bookCode* are foreign keys in the 'branch_has_book' table that reference the 'branch' and 'book' tables, respectively.

3) Table creation script

```
-- 1) Create the database
CREATE DATABASE assign4;

-- 2) Set collation and default character set for db
ALTER DATABASE assign4 CHARSET utf8mb4
                        COLLATE utf8mb4_0900_ai_ci;

-- 3) Set storage engine to be InnoDB
-- Will apply to newly created tables for the new db
SET SESSION default_storage_engine = InnoDB;

-- 4) Activate schema
USE assign4;

/* 5) Create the 'book' table w/:
    bookCode CHAR(4) as the PK
    title VARCHAR(40)
    publisherCode CHAR(3)
    bookType CHAR(3)
    paperback CHAR(1) */

CREATE TABLE book
(
    bookCode CHAR(4) PRIMARY KEY,
    title VARCHAR(40),
    publisherCode CHAR(3),
```

```

        bookType CHAR(3),
        paperback CHAR(1)
    );

/* 6) Create the 'branch' table w/:
    branchNum DECIMAL(2,0) as the PK
    branchName VARCHAR(50)
    branchLocation VARCHAR(50) */
CREATE TABLE branch
(
    branchNum DECIMAL(2,0) PRIMARY KEY,
    branchName VARCHAR(50),
    branchLocation VARCHAR(50)
);

/* 7) Create the 'branch_has_book' table w/:
    branch_branchNum DECIMAL(2,0)
    book_bookCode CHAR(4)
    OnHand DECIMAL(2,0) */
CREATE TABLE branch_has_book
(
    branch_branchNum DECIMAL(2,0),
    book_bookCode CHAR(4),
    OnHand DECIMAL(2,0),
    PRIMARY KEY (branch_branchNum, book_bookCode),
    FOREIGN KEY (branch_branchNum) REFERENCES branch(branchNum),
    FOREIGN KEY (book_bookCode) REFERENCES book(bookCode)
);

```

4a) Insert Records

```

/* 8) Insert 3 records into each table */
INSERT INTO book (bookCode, title, publisherCode, bookType, paperback)
VALUES
('B001', 'Data Science Essentials', 'P01', 'TEC', 'Y'),
('B002', 'Modern SQL Design', 'P02', 'TEC', 'N'),
('B003', 'Marketing Analytics 101', 'P03', 'BUS', 'Y');

INSERT INTO branch (branchNum, branchName, branchLocation)
VALUES
(10, 'Downtown Library', 'Austin'),
(20, 'Campus Library', 'San Marcos'),
(30, 'Westside Library', 'Dallas');

```

```
INSERT INTO branch_has_book (branch_branchNum, book_bookCode, OnHand)
VALUES
```

```
-- Downtown Library carries all three books
```

```
(10, 'B001', 4),
```

```
(10, 'B002', 2),
```

```
(10, 'B003', 5),
```

```
-- Campus Library carries two books
```

```
(20, 'B001', 3),
```

```
(20, 'B002', 6),
```

```
-- Westside Library carries two books
```

```
(30, 'B002', 4),
```

```
(30, 'B003', 3);
```

4b) Retrieve records with an inner join on the tables

```
SELECT
```

```
    br.branchName,
```

```
    br.branchLocation,
```

```
    bk.title,
```

```
    bk.bookType,
```

```
    bhb.OnHand
```

```
FROM branch_has_book AS bhb
```

```
INNER JOIN branch AS br
```

```
    ON bhb.branch_branchNum = br.branchNum
```

```
INNER JOIN book AS bk
```

```
    ON bhb.book_bookCode = bk.bookCode
```

```
ORDER BY br.branchName, bk.bookCode;
```

100%	57:101				
Result Grid		Filter Rows: Search	Export:		
branchName	branchLocati...	title	bookType	OnHand	
Campus Library	San Marcos	Data Science Essentials	TEC	3	
Campus Library	San Marcos	Modern SQL Design	TEC	6	
Downtown Library	Austin	Data Science Essentials	TEC	4	
Downtown Library	Austin	Modern SQL Design	TEC	2	
Downtown Library	Austin	Marketing Analytics 101	BUS	5	
Westside Library	Dallas	Modern SQL Design	TEC	4	
Westside Library	Dallas	Marketing Analytics 101	BUS	3	
Result 3 Read Only					
Action Output					
	Time	Action	Response	Duration / Fetch Time	
1	15:51:18	SELECT br.branchName, br.branchLocation, bk.title, bk.bookType, bhb.OnHand...	7 row(s) returned	0.028 sec / 0.000079...	

